



# SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term Examination (Academic Year 2024-2025)

SY (INFT)

Information Technology

Java Programming

PCC-IT208

Date: 21-06-2025

Time: 02:30 PM to 05:30 PM

Max. Marks: 100

**Instructions:** This question paper consists of 6 questions. All questions are compulsory.

| Q.N  | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks | COs<br>IT20<br>8.X | BT<br>X |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------|---------|
| Q. 1 | a What are static and instance initialization blocks in Java? Explain their purpose, execution order, and use cases with suitable examples. Highlight the difference between a static block and an instance block in terms of when and how they are executed.                                                                                                                                                                                                                                                                                                                                                            | [10]  | 2-3                | 2-4     |
|      | b Explain the concept of inner classes in Java using a suitable example. Demonstrate how an inner class variable can be accessed from the outer class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [10]  | 1-2                | 2-3     |
| Q. 2 | a Explain any 5 String class methods. Provide a suitable example for each and how we can use them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [5]   | 1                  | 2-3     |
|      | b What is inheritance in Java? Describe the types of inheritance supported in Java with appropriate examples. Also, explain the advantages of using inheritance.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [5]   | 3                  | 2-4     |
|      | c Differentiate between String, StringBuffer, and StringBuilder classes in Java regarding mutability, thread safety, and performance. Also, explain the concept of the String Constant Pool/Memory Pool with suitable examples to show how memory is managed when creating string objects.                                                                                                                                                                                                                                                                                                                               | [10]  | 1                  | 2-4     |
| Q. 3 | In what situations can the following errors/exceptions occur in Java programming? Provide a brief example with a short explanation for each of the errors below.                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [12]  | 4                  | 4       |
|      | a Could not find or load main class <i>Sample</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                    |         |
|      | b NullPointerException                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |                    |         |
|      | c ArrayIndexOutOfBoundsException                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                    |         |
| Q. 4 | a Discuss the significance of exception handling in Java for developing robust and error-resilient programs. Additionally, explain the purpose and usage of the try, catch, finally, throw, and throws keywords with appropriate examples.                                                                                                                                                                                                                                                                                                                                                                               | [7]   | 4                  | 1       |
|      | b Write a Java application (either CLI or GUI) that reads content from a text file and prints it on the console. Explain how exception handling is used in file operations.                                                                                                                                                                                                                                                                                                                                                                                                                                              | [8]   | 5-6                | 6       |
| Q. 5 | Provide the output exactly as it would appear when executed in the <b>jshell</b> environment. <b><u>No marks will be awarded without an explanation.</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [15]  | 1-2                | 5       |
|      | a <b>jshell&gt;</b> String.format("%-5x", 25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |                    |         |
|      | b <b>jshell&gt;</b> Integer.toBinaryString(Short.MAX_VALUE)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |                    |         |
|      | c <b>jshell&gt;</b> String s = "Summer Term"<br><b>jshell&gt;</b> s.split(" ")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |                    |         |
|      | d <b>jshell&gt;</b> 34&10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |                    |         |
| Q. 6 | e <b>jshell&gt;</b> int i = 4 > 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | [18]  | 1-2                | 2-5     |
|      | <b>The following Java code contains multiple logical and syntactical errors. Carefully correct all issues so that, when executed, it produces the expected output.</b><br><b>Important Instructions:</b> <ul style="list-style-type: none"> <li>You are <i>not</i> allowed to change the names of any user-defined methods or classes.</li> <li>You may modify existing statements in the code. Complete rewrite of any statement is not allowed, only minimal corrections are permitted.</li> <li>Throughout the code, you are allowed to add only one extra statement (a completely new statement) at most.</li> </ul> |       |                    |         |

```

class Exam {
    String subject;
    int duration;
    {
        System.out.println("All the Best!")
    }

    public Exam(String subject, int duration)
        subject = subject;
        duration = duration;
    }
    void displayDetails() {
        System.out.println("Subject: ",subject ," Duration:
",duration,"hours");
    }
}

class SummerTerm extend Exam {
    String examDate;

    SummerTer(String subject, int duration, String
examDate) {
        subject = subject;
        duration = duration;
        examDate = examDate;
        System.out.print("This is the Summer Term
2024-25!");
    }
    void displayDetails() {
        displayExamDate();
        System.out.println("Exam Date: ", examDate);
    }
}

public class SGGSSummerTermExam {
    public static void main(String args) {
        SummerTerm jpSTE = SummerTerm("Java
Programming", 3, "21st June");
        jpSTE.displayDetails();
    }
}

```

**Expected output:**

*All the Best!*

*This is the Summer Term 2024-25!*

*Subject: Java Programming Duration: 3 hours Exam Date: 21st June 2025*

**Course Outcomes (CO)**

- IT208.1 Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
- IT208.2 Identify classes, objects, members of a class and the relationships among them needed for finding the solution to a specific problem.
- IT208.3 Demonstrates how to achieve reusability using inheritance, interfaces and packages and describes faster application development can be achieved.
- IT208.4 Demonstrate understanding and use of different exception handling mechanisms and concept of multithreading for robust faster and efficient application development.
- IT208.5 Identify and describe common abstract user interface components to design GUI in Java using Applet & AWT along with response to events.
- IT208.6 Identify, Design & develop complex Graphical user interfaces using principal Java Swing classes based on MVC architecture.

**Bloom's Taxonomy (BT)**

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



**Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.**  
**Department of Information Technology**  
**Summer Term Examination**  
**Academic Year: 2024-25**

**Class: S.Y. B.Tech.    Course: Microprocessors and Interfacing    Course Code: PCC-IT207**  
**Date: 19/06/2025                      Time: 14:30 to 17:30                      Max. Marks: 100**

**Note:**

- i. Solve **ANY TEN** questions.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.
- v. Figures to right indicate full marks.
- vi. The symbols carry usual meaning.

| Q. No.   | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark<br>s | CO      | BT        |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|-----------|
| Q.1      | Draw and explain in detail, the block diagram (internal architecture) of Intel 8085 microprocessor.                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [10]      | CO<br>1 | BT1       |
| Q.2      | State the operation performed by the following instructions with the category to which the instruction belongs, number of state and machine cycles required, size of instruction, flags affected and the addressing mode involved:<br>(i) RAL (ii) SHLD addr (iii) ADD M (iv) MVI r, data<br>(v) JMP addr (vi) PUSH rp                                                                                                                                                                                                                                        | [10]      | CO<br>2 | BT3       |
| Q.3      | Draw the timing diagram for the following instructions:<br>(a) INR M (b) ADI data                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [10]      | CO<br>3 | BT2       |
| Q.4      | Write an ALP to arrange a series of <b>ten</b> numbers in Descending order. Write comments for every instruction.                                                                                                                                                                                                                                                                                                                                                                                                                                             | [10]      | CO<br>4 | BT2       |
| Q.5      | Design a memory system for 8085A to have 16 kB ROM and 16 kB RWM. The ROM starts at address 0000H and is implemented using 8 kB EPROMs. EPROM devices have control input (CE) and output enable (OE), both active LOW. The RWM is to be constructed from 4kB SRAMs that have three control lines CE, OE and WE (all active LOW). The SRAM has common I/O and control inputs. Decoding must be exhaustive and must be accomplished using only 74LS138. Draw the logic diagram of the memory interface, memory map and address bit map for the designed system. | [10]      | CO<br>5 | BT6       |
| Q.6 (a)  | Write an ALP to find out 2's compliment of a 16-bit number. Write comment after each instruction.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [05]      | CO<br>3 | BT3,<br>4 |
| Q. 6 (b) | Write an ALP with proper comments to perform sum of series of <b>twelve</b> 8-bit numbers with result 16 bits.                                                                                                                                                                                                                                                                                                                                                                                                                                                | [05]      | CO<br>3 | BT3,<br>4 |
| Q.7      | Draw the block diagram of 8255. Explain in detail control and status words of 8255.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [10]      | CO<br>5 | BT1,<br>2 |
| Q.8 (a)  | Explain memory mapped I/O and I/O mapped I/O with their                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [05]      | CO      | BT2       |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |      |        |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|--------|
|           | advantages and disadvantages.                                                                                                                                                                                                                                                                                                                                                                                                                               |      |      | 1      |
| Q. 8 (b)  | Write a short note on Interrupts of 8085 microprocessor.                                                                                                                                                                                                                                                                                                                                                                                                    | [05] | CO 4 | BT3    |
| Q. 9      | Write an ALP to generate a delay of 400 msec. Assume T = 0.5 $\mu$ sec and give detailed explanation.                                                                                                                                                                                                                                                                                                                                                       | [10] | CO 4 | BT3, 4 |
| Q. 10     | The output device is to be interfaced to the 8085A through port A of 8155. Port B and Port C of 8155 are to be used as input ports. Write a subroutine that initializes 8155 and outputs 100 bytes of data from memory location 7000H of 8085A, to the output device through port A. Assume that 8155 is enabled when A7 to A0 are 01010xxx. Draw the interface diagram. Clearly mention how control word and various port addresses for 8155 are obtained. | [10] | CO 5 | BT4, 6 |
| Q. 11 (a) | Explain various addressing modes and classification of instructions of Intel 8086 microprocessor.                                                                                                                                                                                                                                                                                                                                                           | [05] | CO 1 | BT1, 2 |
| Q. 11 (b) | Explain in detail the flag register and function of various flags available in 8086 microprocessors.                                                                                                                                                                                                                                                                                                                                                        | [05] | CO 1 | BT1    |
| Q. 12 (a) | Write at least 10 differences between Intel's 8085 and 8086 processors.                                                                                                                                                                                                                                                                                                                                                                                     | [05] | CO 2 | BT2    |
| Q. 12 (b) | Write an ALP to add two 8-bit decimal numbers. Write comment after each instruction.                                                                                                                                                                                                                                                                                                                                                                        | [05] | CO 3 | BT3, 4 |

== ALL THE BEST ==

**Bloom's Taxonomy (BT):**

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

|      |                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                            |
| CO 1 | Understand the architecture of 8085 & 8086 microprocessor and understand the difference between them.           |
| CO 2 | Understand and classify the instruction set of 8085 & 8086 microprocessor.                                      |
| CO 3 | Distinguish the use of different instructions of 8085 microprocessor and apply it in Assembly Language Program. |
| CO 4 | Apply the concepts of 8085 programming like interrupts, stacks, subroutines & interfacing.                      |
| CO 5 | Understand and realize the Interfacing of memory & various I/O devices with 8085 microprocessor.                |



Q.N. Solve the followings.

Marks CO BT

Q.1 Explain the concept of error correction in memories. Draw and explain the typical error detection and correction diagram.

10 CO1

OR

Define Computer organization and architecture. How are they different?

Q.2 Write a short note on USB architecture. How does it support plug-and-play and data transfer?

10 CO2

OR

Multiply following pair of signed two's complement represented numbers

A = 0110 B = 1010

Q.3 Given a cache memory of 16 KB with a block size of 4 bytes and a main memory of 1 GB, calculate the number of blocks in the main memory and the number of cache lines. Describe how the blocks are mapped to cache lines using direct mapping.

10 CO3

OR

Solve a disk performance problem calculating access time, latency, and transfer time, given the necessary parameters. Seek Time = 5 ms, Rotational speed = 7200 RPM, Transfer rate = 60 MB/s, Sector size = 512 bytes

CO3

Q.4 Describe and compare the seven levels of RAID configurations with a neat diagram.

OR

10 CO4

Describe the different types of register organizations in a CPU. Explain with suitable diagrams.

Q.5 Discuss about SISD, MIMD, SIMD, SPMD and vector system.

10 CO5

Q.6 Write sign magnitude, two's complement and biased representations for the following numbers in tabular form. -6, -5, 1, 3, 7

10 CO4

Q.7 Explain the role and use of a large register file in RISC architecture. Illustrate with a circular diagram.

10 CO3

Q.8 Design a simple instruction pipeline for a 4-stage pipeline processor. Describe the stages and show how instruction execution is handled in the pipeline.

10 CO5

Q.9 What is a cluster system? Compare Clustered systems to Symmetric Multiprocessing (SMP) systems in terms of structure, performance, and applications.

10 CO6

Q.10 Define superscalar architecture. Explain how super-pipelining is different from traditional pipelining. Mention its limitations (like dependency issues).

#### Course Outcomes (CO)

- CO1. Ability to understand basic structure of computer.
- CO2. Understand the organization of the Control unit, ALU, Memory unit and the I/O unit.
- CO3. Become proficient in quantitative performance evaluation of computer systems
- CO4. Understand how programs and data are stored and represented in a computer system.
- CO5. To be able to design and analyze pipelined control units.
- CO6. To be able to understand and design parallel processing architectures.

#### Bloom's Taxonomy (BT)

BT1-Remember, BT2-Understand, BT3- Apply, BT4-Analyze, BT5-Evaluate, BT6-Create



**Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded**  
**Department of Information Technology**  
**Summer Term Examination**  
**Academic Year: 2024-25**

|                         |                                             |                                    |
|-------------------------|---------------------------------------------|------------------------------------|
| <b>Class: SY</b>        | <b>Subject: Object Oriented Programming</b> | <b>Subject Code: PCC-IT202/203</b> |
| <b>Date: 23.06.2025</b> | <b>Time: 14:30 to 17:30</b>                 | <b>Max. Marks: 100</b>             |

**Note:**

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- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.
- vi. C++ language to be used for code-based examples
- vii. Attempt any 5 questions

| Q.N. | Questions                                                                                                                                                                                | Marks | CO  | BT  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| Q.1  | a Explain following concepts of object-oriented programming in detail with example:<br>1. Classes and Objects<br>2. Encapsulation<br>3. Inheritance<br>4. Polymorphism<br>5. Abstraction | 10    | CO1 | BT1 |
|      | b Explain the following terms:<br>1. Constructor<br>2. Parameterized constructor<br>3. Copy Constructor<br>4. Function overloading<br>5. Operator Overloading                            | 10    | CO1 | BT2 |
| Q.2  | a Explain Call by reference in C++ and differentiate between Pass by Reference and Pass by Pointer.                                                                                      | 10    | CO1 | BT4 |
|      | b Explain Default arguments in a function with example                                                                                                                                   | 10    | CO3 | BT2 |
| Q.3  | a What is exception handling, write a program to illustrate it                                                                                                                           | 10    | CO3 | BT6 |
|      | b What is a constructor, write a program for it                                                                                                                                          | 10    | CO3 | BT6 |
| Q.4  | a What is operator overloading, write a program to overload unary operator                                                                                                               | 10    | CO3 | BT3 |
|      | b Write a program to illustrate single and multiple inheritance                                                                                                                          | 10    | CO3 | BT3 |
| Q.5  | a Explain the following terms with diagram:<br>1. Single Inheritance<br>2. Multiple Inheritance<br>3. Hierarchical Inheritance<br>4. Multilevel inheritance<br>5. Hybrid Inheritance     | 10    | CO3 | BT1 |
|      | b Write a program to read and write a file using "fstream"                                                                                                                               | 10    | CO4 | BT6 |
| Q.6  | a Explain following terms with respect to exception handling                                                                                                                             | 10    | CO2 | BT1 |

|     |                                                                                                                 |    |     |     |
|-----|-----------------------------------------------------------------------------------------------------------------|----|-----|-----|
|     | 1. try keyword<br>2. catch keyword<br>3. throw keyword<br>4. Synchronous Exception<br>5. Asynchronous Exception |    |     |     |
|     | b Illustrate Function template with an example                                                                  | 10 | CO2 | BT2 |
|     | a Explain the concept design patterns in detail                                                                 | 10 | CO5 | BT1 |
| Q.7 | b Explain static keyword, write a program to illustrate static function and static variable                     | 10 | CO3 | BT6 |

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

|      |                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                                                                                           |
| CO 1 | Be able to understand the basics of OOP and Object-oriented approach to design software to solve real world problems based on object-oriented principles                       |
| CO 2 | Be able to test, document and prepare a professional looking software                                                                                                          |
| CO 3 | To be able to apply an object-oriented approach to programming and identify potential benefits of object-oriented programming over other approaches                            |
| CO 4 | Students should be able to read from and write to files using file streams, handle text and binary files, and perform operations such as reading, writing, and appending data. |
| CO 5 | Students should be aware of current trends and advancements in object-oriented programming paradigms, frameworks                                                               |

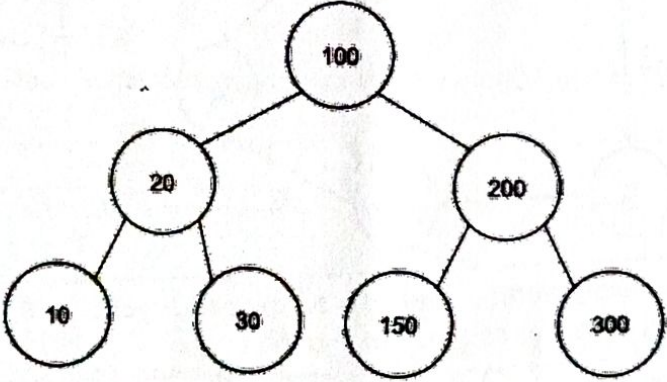


**Shri Guru Gobind Singhji Institute of Engg. and Technology, Vishnupuri, Nanded-431606.**  
**Department of Information Technology**  
**Summer Term Examination**  
**Academic Year: 2024-25**

|                              |                                               |                               |
|------------------------------|-----------------------------------------------|-------------------------------|
| <b>Class: S. Y. B. Tech.</b> | <b>Course: Data Structures and Algorithms</b> | <b>Course Code: MDM-IT201</b> |
| <b>Date: 28/06/2025</b>      | <b>Time: 14.30 to 17.30</b>                   | <b>Max. Marks: 100</b>        |

**Notes:**

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- The symbols carry usual meaning.

| Q. No. | Question                                                                                                                                                                                                     | Marks | CO/BT             |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------|
| Q.1    | a. What is an algorithm? State the characteristics and categories of algorithms.                                                                                                                             | 05    | CO3/<br>BT1, 2    |
|        | b. State the need and classification of data structures. Also differentiate between linear and non-linear data structures.                                                                                   | 05    | CO1/<br>BT1, 2    |
| Q.2    | a. Design a binary expression tree for the given:<br>i. Infix expression: $(A + B * C) - ((D * E + F) / G)$<br>ii. Postfix expression: $A B C * + D /$                                                       | 05    | CO2/<br>BT3, 4, 5 |
|        | b. For the binary tree shown below, print the elements in In-order, Pre-order, Post-order and Level-order traversal.<br> | 05    | CO1/<br>BT3, 4, 5 |
| Q.3    | a. What is a Binary Tree? By drawing suitable diagrams, explain the following binary trees: Full Binary Tree, Complete Binary Tree, and Perfect Binary Tree.                                                 | 05    | CO1/<br>BT1, 2    |
|        | b. State the properties of a Balanced Search Tree. Also state the operations performed on Balanced Search Tree.                                                                                              | 05    | CO2/<br>BT1, 2    |
| Q.4    | Using Selection sort (show stepwise execution), sort the array: [64, 34, 25, 12, 22, 11, 90, 5]. Comment on the time and space complexity of selection sort.                                                 | 10    | CO4/<br>BT4, 5    |
| Q.5    | By taking a simple example, explain the working of Merge sort. Also state its time and space complexity, advantages and disadvantages.                                                                       | 10    | CO4/<br>BT1, 2    |

|      |    |                                                                                                                                                                                                        |    |                   |
|------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------|
| Q.6  |    | For the graph shown below, find the shortest path from the source vertex '0' to all other vertices using <b>Dijkstra's algorithm</b> .                                                                 | 10 | CO5/<br>BT4, 5    |
|      |    | <pre> graph LR     0((0)) --- 4  1((1))     0 --- 8  2((2))     1 --- 6  4((4))     2 --- 2  3((3))     3 --- 10  4 </pre>                                                                             |    |                   |
| Q.7  |    | Apply Binary Search Algorithm to search for the key = 52 from the following data: 4, 7, 12, 16, 27, 34, 39, 46, 52, 68, 75. State the time and space complexity of the Binary search algorithm.        | 10 | CO4/<br>BT3, 4, 5 |
| Q.8  |    | Show the stepwise execution of the Bubble sort algorithm on the data: 56, 22, 4, 8, 29. Comment on the time and space complexity as well as advantages and disadvantages of the Bubble sort algorithm. | 10 | CO3/<br>BT4,5     |
| Q.9  |    | For the graph given below, find the shortest path from the source node '0' to all the other nodes and also find out whether there exists a negative cycle or not using <b>Bellman-Ford Algorithm</b> . | 10 | CO5/<br>BT4,5     |
|      |    | <pre> graph LR     0((0)) -- 5 --&gt; 1((1))     1 -- -2 --&gt; 2((2))     1 -- -3 --&gt; 5((5))     2 -- 3 --&gt; 4((4))     3((3)) -- -2 --&gt; 4     3 -- 1 --&gt; 5     5 -- 6 --&gt; 3 </pre>     |    |                   |
| Q.10 | a. | By drawing a graph, define vertices (nodes), edges, path, cycle, degree of a node, adjacent nodes and loop of a graph.                                                                                 | 05 | CO2/<br>BT1, 2    |
|      | b. | By drawing neat diagrams, explain: Directed graph, Connected graph, Regular graph, Complete graph, and Cycle graph.                                                                                    | 05 | CO1/<br>BT1, 2    |

**//ALL THE BEST//**

**Bloom's Taxonomy (BT):**

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes (COs):**

|      |                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                                  |
| CO 1 | Explain the strengths and weaknesses of different data structures and choose the appropriate one for a specific task. |
| CO 2 | Implement various data structures using a programming language.                                                       |
| CO 3 | Analyze the time and space complexity of algorithms using asymptotic notation.                                        |
| CO 4 | Design efficient algorithms for solving problems using appropriate techniques.                                        |
| CO 5 | Apply data structures and algorithms to solve real-world computing problems.                                          |



**Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded**  
**Department of Information Technology**  
**Summer Term Examination**  
**Academic Year: 2024-25 (Semester – I)**

|                         |                                 |                                 |
|-------------------------|---------------------------------|---------------------------------|
| <b>Class: SY</b>        | <b>Subject: Data Structures</b> | <b>Subject Code: PCC-IT-201</b> |
| <b>Date: 20.06.2025</b> | <b>Time: 14:30 to 17:30</b>     | <b>Max. Marks: 100</b>          |

**Note:**

- i. If in any question asking justification then without correct justification, you will get 0 marks.
- ii. Each Question Carry 10 Marks

| Q. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | C<br>O | B<br>T  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| 1  | Find the location of element in the given array with row major order.<br>a. Loc (X [335]) if array index as X [-150...525], base address is 10000 and size of each element is 10 bytes.<br>b. Loc (Y [175][45]) if array index as Y [-200... 200, -175 ... 75], base address is 100 and size of each element is 5 bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2      | 3,<br>4 |
| 2  | Explain algorithm analysis and the different types of Asymptotic Notations — Big O (O), Omega (Ω), and Theta (Θ) with their trade off diagram with all the bounds.<br>OR<br>Explain Divide and Conquer approach with recursion in details and find the time complexity of Binary Search algorithm using DAC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1      | 2       |
| 3  | Explain stack working and write a program (or pseudocode) to <b>reverse a given string using a stack</b> data structure. Your solution should involve the following steps:<br>1. Push each character of the string onto the stack.<br>2. Pop characters from the stack to build the reversed string.<br><b>Example Input:</b> "ITSA"<br><b>Expected Output:</b> "ASTA"<br>Explain how the stack helps in reversing the string and discuss the time and space complexity of your approach.<br>OR<br>Explain recursion with all its <b>types</b> and solve the below Ackermann Function for A (2,5).<br>$A(m, n) = \begin{cases} n + 1 & \text{if } m = 0 \\ A(m - 1, 1) & \text{if } m > 0 \text{ and } n = 0 \\ A(m - 1, A(m, n - 1)) & \text{if } m > 0 \text{ and } n > 0 \end{cases}$ | 2      | 5       |
| 4  | Implement the circular queue and perform the following operation on the circular queue with size 6 [0-5]. Also show all this operation graphically.<br><b>InsertQ:</b> 11,22,33,44, <b>DeleteQ:</b> 11,22, <b>InsertQ:</b> 55,66,77,88.<br>Write complete InsertQ, DeleteQ, and Display functions for performing the task as the name suggests. Your function must monitor the condition of "Full or Empty."                                                                                                                                                                                                                                                                                                                                                                             | 2      | 2,<br>3 |
| 5  | Design a singly linked list data structure using dynamic memory allocation to store integer values entered by the user. Write pseudocode to perform the following operations on the singly linked list: <b>Insert</b> a new node, <b>delete</b> a node, <b>Sort</b> the linked list in ascending order                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5      | 3,<br>4 |
| 6  | You are given an <b>infix expression</b> that contains integers and standard arithmetic operators. Solve step by step using stack with proper table format.<br>a) Find the <b>postfix expression</b> for the given infix expression: $5 * (6 + 2) - (12 / 4)$<br>b) Do <b>Postfix evaluation</b> for given infix expression: $2 + 3 * 4 / 6 \wedge 1 \wedge 7 + 2 * 3 - 4$                                                                                                                                                                                                                                                                                                                                                                                                               | 2      | 3,<br>5 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 7  | <p>a) Consider the following binary tree data find pre-order <b>tree</b> and pre-order <b>sequence</b>.<br/> Post-order: H J I E F A D G B C      In-order: H G J F I E D A C B</p> <p>b) Consider the following binary search tree (BST) post-order sequence is given. Construct the BST step by step and find the pre-order for that final BST.<br/> Post-order: 10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29</p>                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 | 3, 5 |
| 8  | <p>Write depth-first search traversal (DFT) algorithm and find DFT for starting node A for the given graph. Show all the stack and visited array modification for getting DFT.</p> <div data-bbox="587 456 932 920" data-label="Diagram"> <pre> graph TD     A((A)) --- B((B))     A --- C((C))     A --- D((D))     B --- E((E))     C --- E     C --- F((F))     D --- F     E --- G((G))     F --- G </pre> </div> <p>Also check the given DFT sequence is valid or not for above figure<br/> a) F C A B D R G<br/> b) C E G F D A B<br/> Give the proper justification for the same.</p>                                                                                                                                                                                                                                           | 3 | 3, 4 |
| 9  | <p>Explain hashing with direct address table (DAT) and how the problem of DAT overcome by hash function. Describe the all the types of hash function with an example.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4 | 3, 4 |
| 10 | <p>Consider a hash table of size (<math>m = 10</math>) with the following keys:<br/> 25, 70, 33, 35, 90, 46, 55, 80, 91.<br/> You have to insert all these keys into the hash table, and if there is a collision, then use the following collision resolution techniques with the given hash function:</p> <p>a. Linear Probing [<math>L.P(k, i) = (hf(k) + i) \bmod m</math>]; <math>\forall i \in \{0, 1, 2, \dots, m-1\}</math><br/> where <math>hf(k) = k \bmod m</math></p> <p>b. Quadratic probing [<math>Q.P(k, i) = (hf(k) + c_1 * i + c_2 * i^2) \bmod m</math>];<br/> <math>\forall i \in \{0, 1, 2, \dots, m-1\}</math> where <math>hf(k) = k \bmod m, c_1 = 1, c_2 = 1</math></p> <p>For all the above hash functions, draw the resulting hash table and find how many collisions occur during these keys' insertions.</p> | 4 | 3, 4 |

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

|      |                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                    |
| CO 1 | Understanding the concept of Dynamic memory management, data types, algorithms, Big O notation.         |
| CO 2 | Understand abstract data types, basic data structures such as arrays, linked lists, stacks, and queues. |
| CO 3 | Understand non-linear data structures.                                                                  |
| CO 4 | Describe the hash function and concepts of collision and its resolution methods.                        |
| CO 5 | Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data.           |